

This project will focus on the key OOP principles—Encapsulation, Inheritance, Polymorphism, and Abstraction—and will involve creating a simple Library Management System.

Project: Library Management System

Objectives:

- Practice the basics of **Object-Oriented Programming** (OOP).
- Understand **classes** and **objects**.
- Use **inheritance**, **polymorphism**, **encapsulation**, and **abstraction**.
- Implement **constructors**, **getters and setters**, and **method overloading**.

Project Setup:

- Use **Eclipse IDE**.
 - Create a **Java Console Application** (command-line interface).
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Problem Statement:

You are tasked with creating a simple **Library Management System** that keeps track of books and members. The system should allow library members to **borrow** and **return books**. Implement the basic functionality of the library using OOP principles.

Step-by-Step Guide:

1. Create Classes:

1. **Book** Class:

- Attributes:
 - **bookID** (int)
 - **title** (String)
 - **author** (String)
 - **isAvailable** (boolean)
- Methods:
 - Constructor
 - **getDetails()** – Display book details
 - **borrowBook()** – Change availability status to false
 - **returnBook()** – Change availability status to true

2. **Member** Class:

- Attributes:
 - **memberID** (int)
 - **name** (String)
 - **borrowedBook** (Book)
- Methods:
 - Constructor
 - **borrowBook(Book book)** – Allow member to borrow a book
 - **returnBook()** – Allow member to return the book
 - **getMemberInfo()** – Display member information

3. **Librarian** (extends Member):

- Attributes:
 - **employeeID** (int)
- Methods:

- `addBook(Book book)` – Add a book to the library
 - `removeBook(Book book)` – Remove a book from the library
4. **Library Class:**
- Attributes:
 - `books` (ArrayList of Book)
 - `members` (ArrayList of Member)
 - Methods:
 - `addBook(Book book)` – Add a book to the library
 - `removeBook(Book book)` – Remove a book from the library
 - `getAvailableBooks()` – Display available books
 - `addMember(Member member)` – Add a new member

2. Implement Encapsulation:

- Make class attributes **private** and provide **public getters and setters** for accessing and modifying them.

3. Implement Inheritance:

- The **Librarian** class should inherit from the **Member** class, adding extra functionality for managing books.

4. Implement Polymorphism:

- Use **method overriding** in the **Librarian** class to redefine certain methods if necessary.
- Use **method overloading** for operations like adding or borrowing books.

5. Implement a Menu (Command Line Interface):

- Create a menu for the user (a library member or a librarian) to interact with the system:
 1. **View available books**
 2. **Borrow a book** (by member)
 3. **Return a book** (by member)
 4. **Add a book** (by librarian)
 5. **Remove a book** (by librarian)
 6. **Exit**

Concepts Covered:

- **Encapsulation:** Private attributes with public getters/setters.
- **Inheritance:** Librarian inherits from Member.
- **Polymorphism:** Method overloading/overriding.
- **Abstraction:** Abstract details of book borrowing and returning into methods.